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Naru Jai

LinkedIn
Portfolio

Education

Doctor of Philosophy, Computer Engineering, Virginia Tech, USA **Aug. 2019 – Aug. 2026**
Master of Science, Computer Engineering, Virginia Tech, USA **Aug. 2019 – May 2022**
Bachelor of Engineering, Electronics and Telecommunication, University of Mumbai, India **July 2014 – June 2018**

Skills

Programming	MATLAB, Python, C/C++, Java
Simulation Tools	Jupyter, Spyder, Eclipse
Optimization	Gurobi, CVX
Geospatial Tools	QGIS, Google Earth Pro

Core Technical Competencies

Wireless/RF Systems	5G NR, OFDM, MIMO, beamforming, link budgets, channel modeling
Spectrum Coexistence	CBRS, AFC-enabled 6 GHz access, 3 GHz band, dynamic spectrum sharing, OOB emissions, adjacent-channel leakage, incumbent protection, active–passive coexistence, satellite coexistence
Optimization	Pareto analysis, resource allocation, power control, move lists, mixed-integer optimization
Performance KPIs	RF signal metrics (RSSI, RSRP, RSRQ), link quality (SINR, BER, throughput), coverage, I/N, ACPR
Standards	FCC Part 96, NTIA, WInnForum, 3GPP NR, ITU-R RA.769

Experience

Graduate Research Assistant **Aug. 2020 – May 2026**
Bradley Department of ECE and CCI, Virginia Tech Blacksburg, VA

- Conducted wireless systems research under Prof. Tom Hou and Prof. Yi Shi in the CNSR@VT/Wireless@VT research group, focusing on dynamic spectrum sharing, RF coexistence, and interference management.
- Developed real-time optimization algorithms and simulation models for 5G, CBRS, 6 GHz AFC, and active–passive spectrum-sharing systems, targeting efficient spectrum access under incumbent-protection constraints.
- Designed resource-allocation algorithms for channel assignment, power control, move-list generation, and regulatory-compliance enforcement under FCC, NTIA, 3GPP, ITU, and WInnForum requirements.
- Modeled RF propagation, aggregate interference, out-of-band emissions, and coverage performance using ITM path-loss analysis, GIS-based deployment data, and standards-defined protection thresholds.

Technology Intern **May 2022 – Aug. 2022**
Federated Wireless Arlington, VA

- Researched incumbent protection methods within the CBRS and 3 GHz bands under the supervision of Dr. Masoud Olfat.
- Gained hands-on experience with propagation models and ESC sensors.
- Assisted in preparing critical presentations and technical documents on CBRS band submitted to the FCC and WInnForum.

Teaching Assistant **Jan. 2020 – May 2020**
Virginia Tech Blacksburg, VA

- Instructed laboratory sessions for ECE 3274 – Electronic Circuits Laboratory II, teaching 30 students circuit design, measurement, and data analysis.

Teaching Assistant **Aug. 2019 – Dec. 2019**
Virginia Tech Blacksburg, VA

- Instructed laboratory sessions for ECE 2274 – Electronic Circuits Laboratory I, teaching 30 students with analog circuit design and instrumentation.

Projects

5G Coverage Optimization and RF Coexistence Analysis, **Aug. 2025 – March 2026**

- Developed an interference-constrained RAN optimization algorithm to maximize 5G coverage under 3GPP-defined RSSI thresholds while ensuring that 5G out-of-band emissions into the radio astronomy band remain within regulatory protection limits.
- Built a system-level 5G NR simulation framework that integrates OFDM-based waveform generation, Hamming-window filtering, adjacent-channel leakage analysis, and ITM-based propagation to quantify the trade-off between 5G coverage and interference to radio astronomy receivers, improving coverage by 70% while satisfying 3GPP NR metrics and ITU-R RA.769 protection limits.

Dynamic Spectrum Management for Citizen Broadband Radio Service (CBRS) Band, **Jan. 2025 – July 2025**

- Designed a Pareto-optimal Spectrum Access System (SAS) decision framework to minimize repeated CBRS Devices (CBSDs) service disruptions while enforcing FCC Part 96 incumbent-protection constraints for Navy radar systems in the CBRS band.
- Performed simulation experiments using real-world CBSD locations and NTIA defined Dynamic Protection Areas (DPAs) and showed that the propsed solution reduces repeated CBSD shutdowns by up to 80% relative to baseline approaches while maintaining regulatory-compliant interference limits.

5G RAN Power Control for Incumbent Protection, **May 2024 – Dec. 2024**

- Modeled out-of-band emissions from 5G NR base stations using OFDM-based ACPR and ITM path-loss analysis, and designed a power-control framework to keep aggregate interference at VLBA sites below ITU-R RA.769 limits with minimal BS deactivation.
- Evaluated power-control, move-list, and quiet-zone mechanisms through 5G-realistic simulations, quantifying the coverage–interference trade-off and demonstrating that adaptive power control ensures regulatory compliance while maximizing active base-station density.

Priority-Aware Spectrum Management, **Aug. 2023 – May 2024**

- Formulated a two-stage, mixed-integer SAS optimization framework to minimize CBSD shutdowns during Navy radar activations, while prioritizing Priority Access License (PAL) service continuity over General Authorized Access (GAA) access and satisfying NTIA, FCC Part 96, and CPAS constraints.
- Validated the optimization framework using real DPA geometries and ITM-based propagation models, achieving over 40% reduction in CBSD move-list size while meeting SAS operational constraints and demonstrating readiness for commercial 5G spectrum-management deployment.

Multi-Channel Communications **Aug. 2023 – Dec. 2023**

- *Beamforming*: Designed Max-SNR, MMSE, and LMS transmit/receive beamformers for 4×1 and 1×4 MIMO systems; derived optimal weights and evaluated pilot-assisted channel estimation effects on interference suppression over AWGN and Rayleigh channels.
- *Diversity Techniques*: Implemented MRC, Alamouti STBC, antenna selection, and MRT schemes with pilot-based channel estimation; analyzed BER performance under Rayleigh and Nakagami fading and quantified antenna-correlation effects on diversity gain.
- *MIMO Detection*: Developed ZF, MMSE, MMSE-SIC, and ML detectors with SVD eigenmode beamforming for 2×2 and 4×4 MIMO systems; evaluated BER and performance–complexity trade-offs under channel correlation and CSI constraints.

Reinforcement Learning Algorithm Benchmarking **Aug. 2023 – Dec. 2023**

- Implemented and benchmarked Q-Learning, Actor-Critic, PPO, and DQN algorithms in OpenAI Gym Taxi-v3 to compare convergence speed, training stability, and sample efficiency across tabular and deep reinforcement learning methods.

Modeling and Optimization of Channel Allocation for PAL and GAA Users in the CBRS Band **Nov. 2021 – Dec. 2022**

- Developed a joint channel-allocation framework for PAL and GAA CBSDs that enables real-time spectrum coordination under dynamic incumbent activity, ensuring efficient and interference-aware use of CBRS resources.
- Translated FCC Part 96 and NTIA interference-protection rules into mathematical constraints governing PAL contiguity, GAA coexistence, and DPA/ESC protection, validating the framework with real coastal Virginia CBRS deployments.

Near Real-Time Sensing System for Hydroponics-Based Urban Farming **June 2017 – May 2018**

- Designed and implemented an Arduino-based hydroponic monitoring unit integrating pH, temperature, humidity, light, and moisture sensors for real-time environmental sensing and automated irrigation control.
- Built and tested a working hydroponic prototype with closed-loop water recirculation and coco-peat substrate, demonstrating up to 80% water savings and stable plant growth in constrained indoor environments.

Publications

Journal and Magazine Papers

1. N. Jai, Y. Shi, W. Lou, L. DaSilva, and Y.T. Hou, "Optimizing 5G Coverage under Out-of-Band Interference Protection for Radio Astronomy," *IEEE Transactions on Cognitive Communications and Networking*, under review.
2. N. Jai, Y. Shi, Y. Wu, Y.T. Hou, W. Lou, and L. DaSilva, "Revolving Move Lists: A Novel Approach to Ensure Fairness During Interference Protection Periods in CBRS," *IEEE Transactions on Cognitive Communications and Networking*, under review.
3. N. Jai, Y. Shi, H. Zeng, W. Lou, and Y.T. Hou, "Pay-As-You-Go via Spectrum Bux: A New Paradigm for Dynamic Spectrum Access," *IEEE Communications Magazine*, accepted, 2026.
4. N. Jai, Y. Shi, M. Olfat, J.H. Reed, L. DaSilva, W. Lou, and Y.T. Hou, "An Optimized Move List for Dynamic Protection Area in CBRS," *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 3, pp. 1382–1396, June 2025.
5. N. Jai, Y. Shi, S. Li, C. Li, Y.T. Hou, W. Lou, J.H. Reed, M. Olfat, S. Kompella, and L. DaSilva, "Modeling and Optimization of Channel Allocation for PAL and GAA Users in the CBRS Band," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 1, pp. 1–20, Feb. 2024.

Conference Papers

1. N. Jai, Y. Shi, W. Lou, L. DaSilva, and Y.T. Hou, "Varuna: Maximizing 5G Coverage with Out-of-Band Interference Protection for Radio Astronomy," in *Proc. IEEE MILCOM*, 6 pages, Oct. 6–10, 2025, Los Angeles, CA, USA.
2. N. Jai, Y. Shi, Y. Wu, Y.T. Hou, W. Lou and L. DaSilva, "Taking Turns: Toward Revolving Move Lists for Interference Protection in CBRS," in *Proc. IEEE DySPAN*, 8 pages, May 12–16, 2025, London, UK.
3. N. Jai, Y. Shi, W. Lou, L. DaSilva, and Y.T. Hou, "Out-of-Band Management to Protect Radio Astronomy," in *Proc. IEEE MILCOM*, pp. 1150–1155, Oct. 28–Nov. 1, 2024, Washington D.C., USA. **Vanu Bose Best Paper Award**.
4. N. Jai, Y. Shi, M. Olfat, J.H. Reed, L. DaSilva, W. Lou, and Y.T. Hou, "An Optimized Move List for DPA in CBRS," in *Proc. IEEE DySPAN*, pp. 88-96, May 13–16, 2024, Washington D.C., USA.
5. N. Jai, S. Li, C. Li, Y.T. Hou, W. Lou, J.H. Reed and S. Kompella, "Optimal Channel Allocation in the CBRS Band with Shipborne Radar Incumbents," in *Proc. IEEE DySPAN*, pp. 80-88, Dec. 13–15, 2021, Virtual. **Best Paper Runner-up Award**.
6. N. Jai, B. Dontha, A. Tripathy and S. S. Mande, "Near Real Time - Sensing System for Hydroponics Based Urban Farming," in *Proc. IEEE I2CT*, 5 pages, Apr. 6–8, 2018, Pune, India.

Patent

- **Move Lists for Interference Management in Shared Radio Spectrum**, U.S. Patent Application No. 63/764,924 (Pending).

Awards & Honors

CCI Research Paper Showcase **2025**
- Research paper "Out-of-Band Interference Management to Protect Radio Astronomy" was featured by the Commonwealth Cyber Initiative as impactful CCI-funded research in wireless systems.

Vanu Bose Best Paper Award **2024**
- Awarded by the National Spectrum Consortium at IEEE MILCOM for "Out-of-Band Interference Management to Protect Radio Astronomy," with an associated $5,000 prize.

IEEE MILCOM Student Travel Grant **2024**
- Received a $250 NSF-sponsored student travel grant to attend IEEE MILCOM in Washington, D.C.

IEEE DySPAN Student Travel Grant **2024**
- Received a $1,500 NSF-sponsored student travel grant to attend IEEE DySPAN in Washington, D.C.

CCI SWVA Cyber Innovation Scholar **2023 and 2024**
- Awarded $2,000 professional development grants in both years through the Commonwealth Cyber Initiative SWVA Cyber Scholars Program.

IEEE INFOCOM Student Travel Grant **2023**
- Received a $1,000 NSF-sponsored travel grant to attend IEEE INFOCOM in Hoboken, New Jersey.

Best Poster Award **2022**
- Recognized at the CCI SWVA Student Researcher Showcase for the poster "Robust Interference Mitigation for Spectrum Sharing in the CBRS Band."

Best Paper Runner-up Award **2021**
- Received at IEEE DySPAN for "Optimal Channel Allocation in the CBRS Band with Shipborne Radar Incumbents."

Late Prof. Sundararajan Award **2018**
- Received at INNOVEX, Don Bosco Institute of Technology, for the project "Near Real-Time Sensing System for Hydroponics-Based Urban Farming."

Ranked 3rd in Undergraduate Program **2014–2016**
- Secured 3rd rank during the first two years of the undergraduate program at Don Bosco Institute of Technology in a class of 80 students.

Relevant Courses

- **Wireless & RF:** ECE 5105 – Electromagnetic Waves; ECE 6504 – Spectrum Sharing for 5G and IoT; ECE 6634 – Multi-Channel Communications; ECE 5674 – Software Radios; ECE 5654 – Digital Communication Theory.
- **Signals & Networks:** ECE 5605 – Stochastic Signals and Systems; ECE 6564 – Multimedia Networking; ECE 5565/5566 – Network Architecture and Protocols I–II.
- **Optimization & Algorithms:** ISE 5405/5406 – Optimization I–II; ISE 5984 – Data Analytics and Stochastic Optimization; CS 5040 – Data Structures and Algorithm Analysis.
- **Machine Learning:** ECE 5424 – Advanced Machine Learning; ECE 5984 – Reinforcement Learning.

Presentations

IEEE MILCOM 2025, Los Angeles, CA **Oct. 2025**
- *Paper Title*: Varuna: Maximizing 5G Coverage with Out-of-Band Interference Protection for Radio Astronomy.

IEEE MILCOM 2024, Washington, D.C. **Oct. 2024**
- *Paper Title*: Out-of-Band Interference Management to Protect Radio Astronomy.

IEEE MILCOM Young Scholar Workshop, Washington, D.C. **Oct. 2024**
- *Poster Title*: Interference-Aware Spectrum Sharing for 5G–Passive Service Coexistence.

CCI Symposium 2024, Richmond, VA **April 2024**
- *Poster Title*: Out-of-Band Interference Management to Protect Radio Astronomy.

IEEE DySPAN 2024, Washington, D.C. **May 2024**
- *Paper Title*: An Optimized Move List for DPA in CBRS.

CCI Showcase 2023, Virginia Tech, Blacksburg, VA **March 2023**
- *Poster Title*: Robust Interference Mitigation for Spectrum Sharing in the CBRS Band.

IEEE DySPAN 2021, Virtual Conference **Dec. 2021**
- *Paper Title*: Optimal Channel Allocation in the CBRS Band with Shipborne Radar Incumbents.

Reviewer

- IEEE Transactions on Cognitive Communications and Networking (TCCN).
- IEEE Internet of Things Journal (IoT-J).
- IEEE Transactions on Communications (TCOM).

Professional Memberships

- IEEE Member 2020 – Present
- IEEE Young Professionals (YP) 2020 – Present
- IEEE Communications Society (ComSoc) 2020 – Present
- IEEE Women in Engineering (WIE) 2020 – Present
- IEEE SIGHT 2020 – Present

Leadership & Activities

Web and Graphics Designer, IEEE–DBIT Student Chapter, University of Mumbai **Jan. 2017 – Dec. 2017**
Mumbai, India
- Managed web content and designed promotional materials for IEEE student chapter events.
- Supported event planning, coordination, and outreach initiatives to improve student engagement.

Interests

- Swimming, basketball, reading, and hiking.

References

Dr. Tom Hou
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Dr. Wenjing Lou
W.C. English Endowed Professor of CS
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